

Appendix B: Technical Notes

Determinism I

All stochastic steps use fixed seeds (seed = 42 for NMF, SVD, and graph layouts). The fixture corpus, TF-IDF/SVD embeddings, and topic factorization are byte-stable across runs. Graph centrality scores are rounded to a fixed precision and ranked with a node-id tiebreaker so that floating-point non-associativity in iterative solvers cannot perturb the reported rankings. Record identity uses a content digest (SHA-1, `usedforsecurity=False`) rather than a salted hash, so de-duplication and corpus byte-stability hold across processes.

Data Model I

Each record is a Paper dataclass with: title, abstract, authors (list of Author), year, DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, venue, citation count, references (list of canonical IDs), publication date, PDF URL, open-access flag, and full-text source. The canonical identifier hierarchy governs de-duplication and citation resolution:

$$\text{canonical_id} = \begin{cases} \text{doi:} + \text{normalize(DOI)} & \text{if DOI present} \\ \text{arxiv:} + \text{arXiv_id} & \text{if arXiv ID present} \\ \text{s2:} + \text{S2_id} & \text{if S2 ID present} \\ \text{openalex:} + \text{OpenAlex_id} & \text{if OpenAlex ID present} \\ \text{title:} + \text{SHA1(title)[: 16]} & \text{otherwise} \end{cases}$$

Data Model II

DOI normalization lower-cases the DOI and strips any `https://doi.org/` or `dx.doi.org/` prefix, so the same paper returned by two engines under different case or format variants merges to a single canonical ID.

NMF Mathematics I

Non-negative matrix factorization decomposes the TF-IDF document-term matrix $\mathbf{V} \in \mathbb{R}^{m \times n}$ (where m is the number of documents and n is the vocabulary size) into $\mathbf{W} \in \mathbb{R}^{m \times k}$ and $\mathbf{H} \in \mathbb{R}^{k \times n}$, where k is the number of topics (here 5). The factorization minimizes:

$$\min_{\mathbf{W}, \mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{WH}\|_F^2$$

using multiplicative update rules [lee1999learning] with a fixed random seed for reproducibility. The topic-term matrix $\mathbf{H}$ gives the top terms per topic; the document-topic matrix $\mathbf{W}$ gives each document's topic loadings.

Growth Rate Estimation I

The compound annual growth rate is:

$$\text{CAGR} = \left(\frac{N_{\text{end}}}{N_{\text{start}}} \right)^{1/(T_{\text{end}} - T_{\text{start}})} - 1$$

where N_{start} and N_{end} are the publication counts in the first and last years of the corpus, respectively. The doubling time is $t_d = \ln(2)/\ln(1 + \text{CAGR})$. For this run: $\text{CAGR} = 12.27\%$, doubling time = 1.4 years.

Configuration Surface I

A single `manuscript/config.yaml` controls the search term, per-engine query and keyword sets, engine enable toggles, subfield taxonomy, hypotheses, full-text and embedding options, and paper metadata. This run drew on 4 engines, a 4-bucket taxonomy, and 4 hypotheses.

Artifacts I

Intermediate and final outputs live under output/ and are disposable and regenerable:

File	Stage	Description
corpus.jsonl	01	De-duplicated corpus (46 records)
temporal_analysis.json	02	Year counts, CAGR, doubling time, peak year
subfield_classification.json	02	Per-bucket paper counts
subfield_timeline.json	02	Per-subfield annual breakdown
tfidf_data.json	02	TF-IDF matrix, feature names, doc tokens
topics.json	02	NMF topic-term distributions
citation_network.json	02	Network metrics, PageRank, HITS, communities
citation_graph.gml	02	GraphML citation graph
nanopublications.jsonl	03	LLM-extracted assertions (0 in this run)
hypothesis_scores.json	03	Per-hypothesis evidence scores
cross_phase_analysis.json	01 / 03	Phase membership, overlap, citation sufficiency, and explicit hypothesis-score claim boundary
fulltext_assessment.json	06	Abstract/OA/PDF coverage report